

Importance Sampling

Monte Carlo Statistical Methods

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1 Motivation: Two Dice and the Birth of Weights

Suppose we want to understand the behavior of a **biased die** whose outcomes follow a certain probability distribution $f(x)$ on $\{1, 2, 3, 4, 5, 6\}$. However, we cannot roll this die directly. Instead, we only have access to a **fair die** that follows a simpler distribution $g(x)$, from which we can easily roll samples.

The question is: if we can only roll the fair die, how can we make those samples behave statistically as if they were drawn from the biased one? This simple question captures the essence of *importance sampling*.



Target

(Biased) Die $f(x)$						
x	1	2	3	4	5	6
$f(x)$	0.25	0.25	0.15	0.15	0.10	0.10



Proposal

(Fair) Die $g(x)$						
x	1	2	3	4	5	6
$g(x)$	1/6	1/6	1/6	1/6	1/6	1/6

Question 1: If we roll the fair die $g(x)$ 600 times, how many 1s and 6s do we expect?

$$\mathbb{E}[\#(x)] = 600 \times g(x) \Rightarrow \text{1s: } 600 \times \frac{1}{6} = 100, \quad \text{6s: } 600 \times \frac{1}{6} = 100.$$

So we expect about **100 ones** and **100 sixes**.

Question 2: If we could roll the biased die $f(x)$ 600 times, how many 1s and 6s would we expect?

$$\mathbb{E}[\#(x)] = 600 \times f(x) \Rightarrow \text{1s: } 600 \times 0.25 = 150, \quad \text{6s: } 600 \times 0.10 = 60.$$

So we would expect **150 ones** and only **60 sixes**.

The challenge. We cannot roll the biased die $f(x)$, but we can roll the fair die $g(x)$. After 600 rolls, $g(x)$ gives ≈ 100 of each face, while $f(x)$ requires more 1s and fewer 6s. *How can we make our fair die samples behave as if they came from $f(x)$?*

The natural fix: scale the counts.

- Suppose we roll $g(x)$ 600 times and get ≈ 100 outcomes for each face.
- Under $f(x)$ there should be 150 ones and only 60 sixes.
- *Reinterpret each observation with a weight (multiplier) for how much it should count.*

If g gave 100 ones but f expects 150, then each “1” from g should count 1.5 times more:

$$w(1) = \frac{f(1)}{g(1)} = 1.5.$$

If you rolled 10 ones from the fair die, you treat them as representing 15 ones in the biased world. Those 10 rolls now carry the same statistical weight as 15 biased-die rolls.

Similarly, if g gave 100 sixes but f expects only 60, then each “6” should count less:

$$w(6) = \frac{f(6)}{g(6)} = 0.6.$$

If you observed 10 sixes from the fair die, you treat them as representing only 6 sixes under f . They still exist; they just contribute less to the final estimate.

This gives rise to the **importance weight**:

$$w(x) = \frac{f(x)}{g(x)}.$$

x	1	2	3	4	5	6
$f(x)$	0.25	0.25	0.15	0.15	0.10	0.10
$g(x)$	1/6	1/6	1/6	1/6	1/6	1/6
$w(x) = f(x)/g(x)$	1.5	1.5	0.9	0.9	0.6	0.6

Weights correct the sampling frequency difference between f and g .

Interpreting the weights.

- Weights do not change the face rolled; they change *how much that roll matters*.
- Each observation x_i from $g(x)$ is viewed as standing in for $w(x_i)$ “virtual” observations from $f(x)$.
- Large weights (e.g., 1.5) emphasize under-sampled outcomes; small weights (e.g., 0.6) down-weight over-sampled ones.

For example:

- If $x_i = 1$, we count it as $w(1) = 1.5$ rolls of “1”.
- If $x_i = 6$, we count it as $w(6) = 0.6$ rolls of “6”.

In this way, the weighted collection of rolls from g statistically mirrors what would have been produced by sampling directly from f . The weights translate

“how often we actually saw x ” $\rightarrow$ “how often we should have seen x under f ”.

From counts to statistics. The same correction applies to any statistic. Given rolls $x_1, \dots, x_n \sim g(x)$,

$$\mathbb{E}_f[h(X)] = \sum_x h(x)f(x) = \sum_x h(x)\frac{f(x)}{g(x)}g(x) = \mathbb{E}_g[h(X)w(X)] \approx \frac{1}{n} \sum_{i=1}^n h(x_i)w(x_i).$$

Finite-sample drift and self-normalization (preview). With finite n , empirical counts fluctuate, so $\sum_i w(x_i)$ may differ from n . To keep the total mass consistent, we define normalized weights

$$\tilde{w}_i = \frac{w(x_i)}{\sum_{j=1}^n w(x_j)}, \quad \sum_{i=1}^n \tilde{w}_i = 1,$$

and use the weighted average $\sum_i \tilde{w}_i h(x_i)$. These *self-normalized weights* ensure that the total probability mass remains constant even with sampling noise.

Key idea.

$$w(x) = \frac{f(x)}{g(x)}$$

is the scaling factor that transforms “how often we actually rolled something” under g into “how often we should have rolled it” under f . Larger weights amplify underrepresented outcomes; smaller weights suppress overrepresented ones.